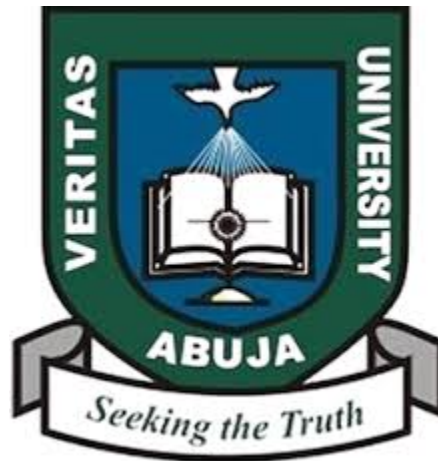


Department of Software Engineering
Veritas University, Bwari Abuja.



‘LAB MANUAL’
Course Code: VUA-SEN182
Course Title: “Web Programming (HTML, CSS, Git)”

Prepared by: Mr. Victor Ayamba

Course Contents

Contents	CLOs	Lecture	Practical
HTML: Basics, Elements, Attributes, Headings, Paragraphs, Formatting, Links, Head, Images, Tables, Lists, Blocks, Layout, Forms, Multimedia Tags, Semantic HTML, HTML5 Structure.	CLO – 1 & 2	4	8
CSS: Introduction, Syntax, Selectors (ID & Class), Styling Backgrounds, Text, Fonts, Links, Box Model (Border, Outline, Margin, Padding), Grouping/Nesting, Dimension, Display, Positioning, Floating, Align, Navigation Bar, Image Gallery, Flexbox, Grid Layout, Responsive Design.	CLO – 1 & 2	3	6
JavaScript: Introduction, Output, Statements, Comments, Variables, Data Types, Objects, Functions, Operators, Comparisons, Conditions, Loops, Error Handling, DOM Introduction and Manipulation, Number, String, Date, Array, Window, Screen, Navigator, Popup Alerts.	CLO – 1 & 2	4	8
Bootstrap: Introduction to Bootstrap Framework, Grid System, Containers and Layout, Responsive Design Principles, Typography and Utilities, Navigation Bars, Buttons, Cards, Forms and Input Groups, Tables, Alerts, Modals, Responsive Components and Layout Customization.	CLO – 1 & 2	4	8
Total		15	30

INTRODUCTION TO THE COURSE:

The Internet and the World Wide Web have become fundamental components of modern society, influencing communication, education, business, and everyday life. As digital platforms continue to expand, the ability to design and develop functional and visually appealing web applications has become an essential skill for software engineers and technology professionals. Understanding how web technologies work together enables students to create interactive systems that can serve real-world needs.

This laboratory course provides a practical introduction to modern Web Programming using core client-side technologies such as **HTML, CSS, JavaScript, and the Bootstrap framework as well as version control tools like Git and Github**. Students will learn how to structure web pages using HTML, style and layout interfaces with CSS, enhance interactivity through JavaScript, and build responsive, mobile-friendly user interfaces using Bootstrap. Through hands-on laboratory exercises, students will progressively develop the skills required to design, implement, and maintain modern web applications.

The lab sessions are structured to guide students step-by-step through the development of a meaningful web project while introducing essential web development concepts and tools. Each experiment builds upon the previous one, enabling learners to gradually transform basic web pages into fully structured and responsive websites. At the end of each lab session, tasks are provided to reinforce understanding and encourage independent exploration. Students are advised to follow this lab manual sequentially and carefully adhere to the provided guidelines to gain maximum benefit from the course.

Lab 4: HTML Forms and Input Elements - Demonstrated by Instructor

HTML Forms

HTML forms are used to collect user input. User input can include text, numbers, email addresses, passwords, selections, and other information. Forms are very important in web development because they allow websites to receive information from users.

An HTML form is created using the `<form>` tag.

Example:

```
<form>  
</form>
```

Inside the form, different **input elements** are used to collect different kinds of data.

Common HTML Form Elements

The main HTML form elements include:

- `<form>`
- `<input>`
- `<label>`
- `<textarea>`
- `<select>`
- `<option>`
- `<button>`
- `<fieldset>`
- `<legend>`

The `<form>` Element

The `<form>` element is used to create an HTML form for user input.

Example:

```
<form>
```

```
<label for="fullname">Full Name:</label>
<input type="text" id="fullname" name="fullname">
</form>
```

The **<input>** Element

The **<input>** element is the most commonly used form element.

The **type** attribute specifies the kind of input field to display.

Example:

```
<input type="text">
```

HTML Input Types

HTML provides different kinds of input types for different purposes.

Common Input Types:

- text
- password
- email
- number
- radio
- checkbox
- date
- file
- submit
- reset
- button

Text Input

Text input is used to collect single-line text such as names.

Example:

```
<label for="fname">First Name:</label>
<input type="text" id="fname" name="fname">
```

Password Input

Password input hides the characters entered by the user.

Example:

```
<label for="password">Password:</label>
<input type="password" id="password" name="password">
```

Email Input

Email input is used for collecting email addresses.

Example:

```
<label for="email">Email:</label>
<input type="email" id="email" name="email">
```

Number Input

Number input is used when the user is expected to enter numeric values only.

Example:

```
<label for="age">Age:</label>
<input type="number" id="age" name="age">
```

Radio Buttons

Radio buttons allow the user to choose **one option only** from a group.

Example:

```
<p>Gender:</p>
<input type="radio" id="male" name="gender" value="Male">
<label for="male">Male</label>
<input type="radio" id="female" name="gender" value="Female">
<label for="female">Female</label>
```

Checkboxes

Checkboxes allow the user to select **multiple options**.

Example:

```
<p>Select your interests:</p>
<input type="checkbox" id="sports" name="interest" value="Sports">
<label for="sports">Sports</label>
<input type="checkbox" id="music" name="interest" value="Music">
<label for="music">Music</label>
```

Date Input

Date input allows the user to select a date.

Example:

```
<label for="dob">Date of Birth:</label>
<input type="date" id="dob" name="dob">
```

File Input

File input allows users to upload files.

Example:

```
<label for="passport">Upload Passport:</label>
<input type="file" id="passport" name="passport">
```

Submit Input

Submit input is used to submit the form data.

Example:

```
<input type="submit" value="Submit">
```

Reset Input

Reset input clears all form fields.

Example:

```
<input type="reset" value="Reset">
```

Button Input

A button can be used for custom actions.

Example:

```
<input type="button" value="Click Me">
```

The <label> Element

The <label> element defines a label for an input field. It improves usability because clicking the label activates the related input.

Example:

```
<label for="fullname">Full Name:</label>
<input type="text" id="fullname" name="fullname">
```

The <textarea> Element

The <textarea> element is used to collect multi-line text such as comments, feedback, or addresses.

Example:

```
<label for="message">Message:</label>
<textarea id="message" name="message" rows="4" cols="30"></textarea>
```

The <select> and <option> Elements

The <select> element creates a dropdown list, while <option> defines the selectable options.

Example:

```
<label for="faculty">Faculty:</label>
<select id="faculty" name="faculty">
  <option value="science">Science</option>
  <option value="arts">Arts</option>
  <option value="management">Management Sciences</option>
</select>
```

The `<fieldset>` and `<legend>` Elements

The `<fieldset>` element groups related form elements together, while `<legend>` provides a caption for the group.

Example:

```
<fieldset>
  <legend>Personal Information</legend>

  <label for="fullname">Full Name:</label>
  <input type="text" id="fullname" name="fullname">
</fieldset>
```

Form Attributes

Some common form attributes include:

- **action** : Specifies where the form data should be sent
- **method** : Specifies how form data should be sent (**get** or **post**)
- **required** : Makes a field compulsory
- **placeholder** : Displays a short hint inside an input field
- **value** : Sets a default value

Example:

```
<form action="/submit" method="post">
  <input type="text" name="fullname" placeholder="Enter your full name"
  required>
</form>
```

Example of a Complete Form

```
<form>
  <fieldset>
    <legend>Student Orientation Registration</legend>

    <label for="fullname">Full Name:</label>
    <input type="text" id="fullname" name="fullname"><br><br>
```

```
<label for="email">Email:</label>
<input type="email" id="email" name="email"><br><br>

<label for="age">Age:</label>
<input type="number" id="age" name="age"><br><br>

<p>Gender:</p>
<input type="radio" id="male" name="gender" value="Male">
<label for="male">Male</label>

<input type="radio" id="female" name="gender" value="Female">
<label for="female">Female</label><br><br>

<p>Documents Submitted:</p>
  <input type="checkbox" id="admission" name="docs" value="Admission
Letter">
  <label for="admission">Admission Letter</label>

  <input type="checkbox" id="receipt" name="docs" value="Payment
Receipt">
  <label for="receipt">Payment Receipt</label><br><br>

<label for="faculty">Faculty:</label>
<select id="faculty" name="faculty">
  <option value="science">Science</option>
  <option value="arts">Arts</option>
  <option value="management">Management Sciences</option>
</select><br><br>

<label for="message">Additional Information:</label><br>
  <textarea id="message" name="message" rows="4"
cols="40"></textarea><br><br>

  <input type="submit" value="Submit">
  <input type="reset" value="Reset">
</fieldset>
</form>
```

Student Orientation Registration

Full Name:

Email:

Age:

Gender:

☐ Male ☐ Female

Documents Submitted:

☐ Admission Letter ☐ Payment Receipt

Faculty:

Additional Information:

Figure 1 : Complete Form

SECTION B: Lab Tasks on HTML Forms and Input Elements

Task 1: Create a Basic Form

What to do

Create a basic HTML form for the Veritas University Orientation Portal.

Suggestion

Use the `<form>` tag as the main container.

Task 2: Add Text Input Fields

What to do

Add text input fields to collect the student's **Full Name** and **Matriculation Number**.

Suggestion

Use `<label>` and `<input type="text">`.

Task 3: Add an Email Input

What to do

Add a field to collect the student's email address.

Suggestion

Use `<input type="email">`.

Task 4: Add a Number Input

What to do

Add a field to collect the student's age.

Suggestion

Use `<input type="number">`.

Task 5: Add Radio Buttons

What to do

Add radio buttons to allow the student select **gender**.

Suggestion

Task 6: Add Checkboxes

What to do

Add checkboxes for students to indicate the documents they have brought for registration.

Suggestion

Use `<input type="checkbox">`.

Task 7: Add a Date Input

What to do

Add a field for the student to select their date of birth.

Suggestion

Use `<input type="date">`.

Task 8: Add a File Upload Input

What to do

Allow students to upload a passport photograph or admission document.

Suggestion

Use `<input type="file">`.

Task 9: Add a Dropdown List

What to do

Add a dropdown list for students to select their faculty.

Suggestion

Use `<select>` and `<option>`.

Task 10: Add a Text Area

What to do

Add a text area where students can enter additional comments or questions.

Suggestion

Use `<textarea>`.

Task 11: Group the Form Elements

What to do

Group all related input fields into one form section.

Suggestion

Use `<fieldset>` and `<legend>`.

Task 12: Add Submit and Reset Buttons

What to do

Add buttons to submit and clear the form.

Suggestion

Use `<input type="submit">` and `<input type="reset">`.

Task 13: Add Useful Form Attributes

What to do

Improve the form by making some fields required and adding placeholders.

Suggestion

Use `required` and `placeholder`.

Task 14: Build the Full Orientation Registration Form

What to do

Combine all the form elements you have learned into one **Student Orientation Registration Form** for the Veritas University Orientation Portal.

Suggestion

Your form should collect:

- Full Name
- Matric Number
- Email
- Age
- Gender
- Date of Birth
- Faculty
- Documents Submitted
- Passport Upload
- Additional Comments